SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	Gangisetty Anjani Kumar	**Reg. No.:**	192373022
Section / Batch:	CSE(DS)	**Date:**	07-10-2026

Code of Conduct

I _G. Anjani Kumar (192373022)___ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

*Signature of the Student: _G. Anjani kumar_*

Instructions

- This test contains ONE question.
- Duration: 30 minutes (At home)

Annexure A - QUESTIONS

- Develop a concept map that evaluates the relationship between mobility, handoff, and roaming. Illustrate their dependencies and interactions. Determine which mechanism contributes most to uninterrupted communication.
  Justify your evaluation using appropriate linking phrases.

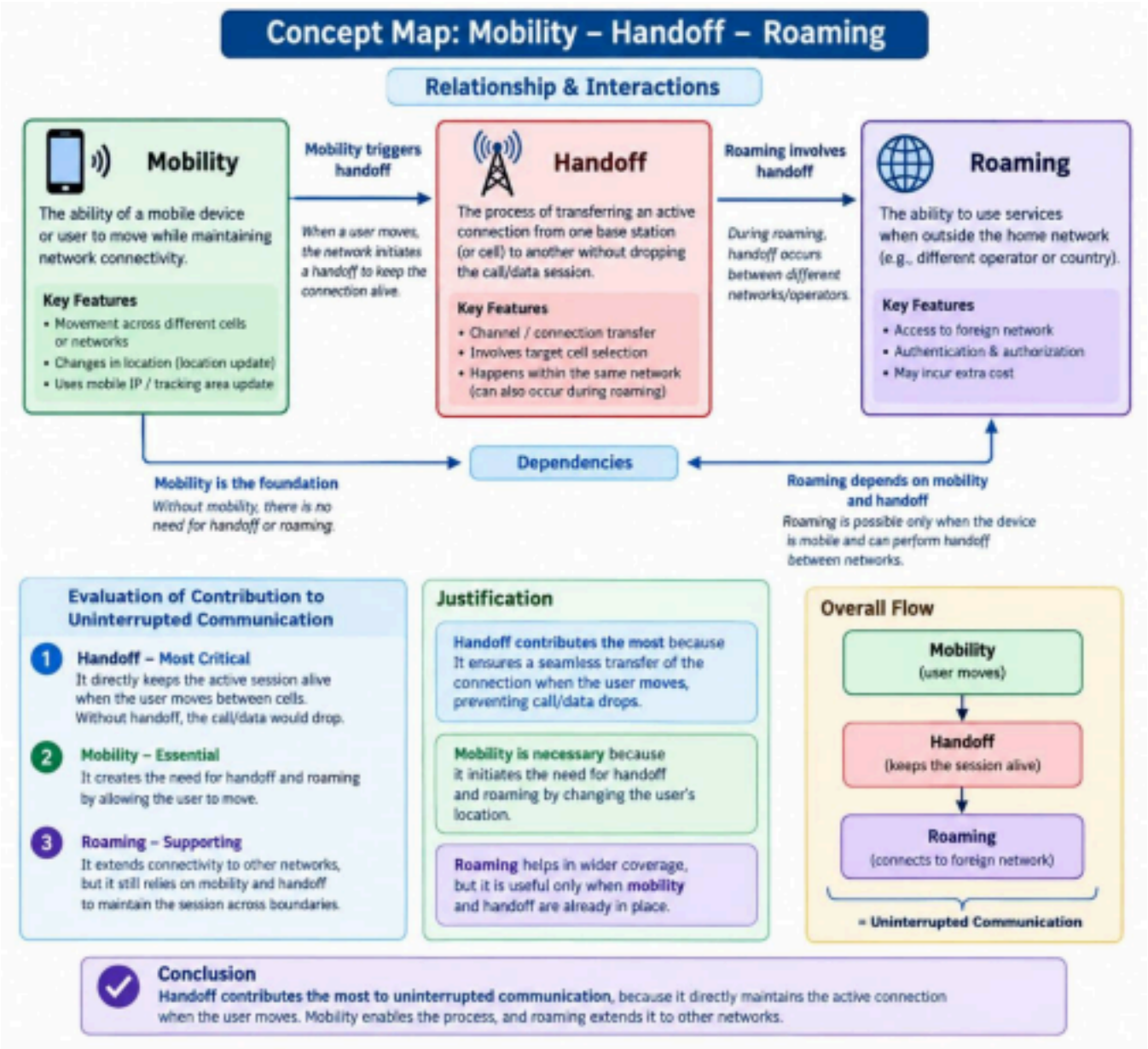
Concept map points: Mobility, Handoff and Roaming

- Mobility → *means* → users move while accessing a mobile network. • Mobility → *creates the need for* → handoff when users move between cell coverage areas.
- Handoff → *transfers* → an active call or data connection from one cell/base station to another.
- Handoff → *maintains* → connection continuity and reduces call drops. • Mobility

across operator coverage boundaries → *creates the need for* → roaming. • Roaming → *allows* → users to access services through a visited network outside their home network.

- Roaming → *depends on* → operator agreements, authentication and subscriber authorization.
- Roaming → *works with* → handoff to support users moving between cells within a visited network.
- Handoff and roaming → *together support* → communication during mobility.

Evaluation: Handoff contributes most directly to uninterrupted communication because it preserves an ongoing connection while the user moves between cells. Roaming mainly ensures service availability outside the home network; it does not, by itself, guarantee that an active call continues without interruption.



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem Solving Approach	20	Logical, well structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____